1. Clinical Question & Core Hypothesis

Clinical Question

PAM50-defined Luminal A tumors (transcriptomically ER-high, low-proliferation) are expected to be endocrine-sensitive, yet a subset still exhibits primary or early acquired resistance — what microenvironmental features convert an otherwise endocrine■sensitive program into clinical nonresponse? 2) Tier 1 paths used: (a) Pretreatment transcriptome → PAM50 → Luminal A → endocrine therapy response (mechanistic backbone linking ESR1-driven hormone signaling and low proliferation to endocrine sensitivity); (b) Pretreatment transcriptome → proliferation gene signature (MKI67 et al.) → endocrine response (proliferation modulates sensitivity). I built on these because they directly explain why tumors classified as Luminal A may or may not respond. 3) Tier 2 evidence supporting the claim: stromal/ECM programs (NABA_CORE_MATRISOME genes such as FN1, COL1A1), and growth■factor paracrine signaling from stroma can sustain estrogen■independent proliferation — these provide a plausible mechanism by which high stromal-to-tumor ratio (TSR) blunts endocrine efficacy. 4) Why therapy_response endpoint: the translational question is explicitly about endocrine benefit (prediction and mechanism of endocrine resistance), so a therapy_response endpoint directly tests the clinical consequence of the mechanistic interaction. 5) Deliberate exclusions: I excluded immune-organizing features (TIL density, TLS) to keep focus on tumor–stroma interactions; I limited core molecular anchors to ESR1 and MKI67 to maintain the 2-gene maximum and relegated other proliferation and ECM genes (CCNB1, BIRC5, AURKA, FN1, COL1A1) to secondary_modifiers for follow-up analyses?

Working Hypothesis

In HR+/HER2− breast cancers classified as PAM50 Luminal A (characterized by high ESR1 expression and low MKI67), a high stromal-to-tumor ratio (TSR) independently predicts reduced response to endocrine therapy. The proposed mechanism is that TSR-rich stroma enhances extracellular matrix and paracrine signaling (driven by ECM genes such as FN1 and COL1A1) that activate alternative pro-proliferative pathways, enabling estrogen-independent tumor cell growth despite ER-targeted therapy. ESR1-driven transcription remains active but is functionally overridden by stromal cues in TSR-high tumors, diminishing endocrine efficacy. MKI67 serves as a secondary modifier, where higher baseline proliferation within Luminal A tumors may amplify resistance in the TSR-high context. Clinically, Luminal A tumors with high TSR will show lower therapy_response rates than TSR-low counterparts, even after adjusting for ESR1 expression and MKI67 level. This hypothesis implies ESR1 and MKI67 are necessary but not sufficient determinants of response; the TSR context modulates therapeutic outcome through microenvironmental signaling that sustains proliferation independent of estrogen signaling.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	ESR1, MKI67
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

ESR1: ESR1 and MKI67 to maintain the 2-gene maximum and relegated other proliferation and ECM genes (CCNB1, BIRC5, AURKA, FN1, COL1A1) to secondary_modifiers for follow-up analyses.

MKI67: MKI67 to maintain the 2-gene maximum and relegated other proliferation and ECM genes (CCNB1, BIRC5, AURKA, FN1, COL1A1) to secondary_modifiers for follow-up analyses.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CCNB1	Supporting modifier from knowledge graph
Secondary	BIRC5	Supporting modifier from knowledge graph
Secondary	AURKA	Supporting modifier from knowledge graph
Secondary	FN1	Supporting modifier from knowledge graph
Secondary	COL1A1	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: , are typically endocrine-sensitive. However, the tumor microenvironment—particularly stromal abundance and ECM remodeling—can provide alternative growth signals that partially sustain proliferation when estrogen signaling is inhibited, potentially causing primary or early acquired resistance.

Prediction 2: retrospective analyses of PAM50-classified cohorts with quantified TSR (histology or gene-signature-based proxy) and documented endocrine therapy outcomes, testing TSR as an independent predictor after adjusting for ESR1 and MKI67;

Prediction 3: in vitro co-culture models of Luminal A–like breast cancer cell lines with cancer-associated fibroblasts to mimic high TSR; assess endocrine therapy response (proliferation, apoptosis) with and without blockade of stromal signaling pathways (e.g., FGFR, IGF1R, HGF/c-MET) or ECM remodeling;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CCNB1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- BIRC5: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.